# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

# MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

## GROUP – I

### Mathematics (Objective) Group - I

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 7193

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. How many common tangents can be drawn for two touching circles?

(A) 2 (B) 3

(C) 4 (D) 5

دو مس کرت ہوئے دائروں کے کتنے مشترک مماس بنائے جا سکتے ہیں؟

Q2. The length of a chord and the radial segment of a circle are congruent, the central angle made by the chord will be:

(A) 30° (B) 45°

(C) 60° (D) 75°

ایک دائرے میں وتر اور اس کی لمبائیں مماثل ہیں تو وتر سے بننے والا مرکزی زاویہ ہوگا:

Q3. A circle has only one:

(A) Secant
(B) Chord
(C) Diameter
(D) Center

ایک دائرے کا صرف ایک _____ ہوتا ہے۔

**(A)** خط قاطع
**(B)** وتر
**(C)** قطر
**(D)** مرکز

Q4. Locus of a point in a plane equidistant from a fixed point is called:

(A) Circle
(B) Radius
(C) Diameter
(D) Circumference

مستوی کے تمام نقاط کا سیٹ جو معین نقطے سے برابر فاصلے پر ہوں کہلاتا ہے:

**(A)** دائرہ
**(B)** رداس
**(C)** قطر
**(D)** محیط

Q5. sec θ cot θ = _____

(A) sin θ
(B) 1/cos θ
(C) sin θ/cos θ
(D) 1/sin θ

Q6. Mean is affected by change in:

(A) Place
(B) Scale
(C) Rate
(D) Number

حسابی اوسط _____ تبدیل کرنے سے اثر انداز ہوتا ہے۔

**(A)** جگہ
**(B)** پیمانہ
**(C)** مقدار/شرح
**(D)** نمبر

Q7. A histogram is a set of adjacent:

(A) Squares
(B) Rectangles
(C) Circles
(D) Rows

ہسٹوگرام ملحقہ _____ کا مجموعہ ہے۔

**(A)** مربعوں کا
**(B)** مستطیلوں کا
**(C)** دائروں کا
**(D)** قطاروں کا

Q8. The range of R = {(1, 3), (2, 2), (3, 1), (4, 4)} is:

(A) {1, 2, 4}
(B) {3, 2, 4}
(C) {1, 2, 3, 4}
(D) {1, 3, 4}

اگر $R=\{(1,3),(2,2),(3,1),(4,4)\}$ ہو تو range R ہوتی ہے:

Q9. A set with no element is called:

(A) Sub set
(B) Empty set
(C) Singleton set
(D) Super set

ایسا سیٹ جس میں کوئی رکن نہ ہو کہلاتا ہے:

**(A)** تحتی سیٹ
**(B)** خالی سیٹ
**(C)** یک رکنی سیٹ
**(D)** سپر سیٹ

Q10. A fraction in which the degree of numerator is less than the degree of the denominator is called:

(A) An equation
(B) An improper fraction
(C) An identity
(D) A proper fraction

کسر جس میں شمار کنندہ کی ڈگری مخرج کی ڈگری سے کم ہو کہلاتی ہے:

**(A)** مساوات
**(B)** غیر واجب کسر
**(C)** مماثلت
**(D)** واجب کسر

Q11. If a : b = x : y, then invertendo property is:

(A) a/x = b/y
(B) a/b = x/(x-y)
(C) (a+b)/b = (x+y)/y
(D) b/a = y/x

اگر $a:b=x:y$ ہو تو معکوس نسبت ہے:

Q12. If $y^2 \propto 1/x^3$, then:

(A) $y^2 = k/x^3$
(B) $y^2 = 1/x^2$
(C) $y^2 = x^2$
(D) $y^2 = kx^3$

اگر $y^2 \propto 1/x^3$ ہو تو:

Q13. Sum of cube roots of unity is:

(A) 0
(B) 1
(C) -1
(D) 3

اکائی کے جزر المکعب کا مجموعہ ہے:

Q14. If α, β are the roots of $px^2 + qx + r = 0$, then sum of the roots 2α and 2β is:

(A) q/p
(B) r/p
(C) -2q/p
(D) -q/2p

اگر α, β مساوات $px^2 + qx + r = 0$ کے روٹس ہوں تو 2α اور 2β کا مجموعہ ہے:

Q15. The number of methods to solve a quadratic equation is:

(A) 1
(B) 2
(C) 3
(D) 4

دو درجی مساوات کو حل کرنے کے کتنے طریقے ہیں؟

## Mathematics (Subjective) Group - I

**Time Allowed:** 02:10 Hours | **Maximum Marks:** 60 | **Paper Code:** 7193

Note: Part - I (حصہ اول)

Q2. (Marks 12) Write short answers to any SIX parts.

(i) Write quadratic equation in standard form: (x + 7)(x - 3) = -7

(ii) Write the names of the methods for a quadratic equation.

(iii) Solve by factorization: $3y^2 = y(y - 5)$

(iv) Find the discriminant of given quadratic equation: $2x^2 + 3x - 1 = 0$

(v) Without solving, find the sum and the product of the roots of the quadratic equation: $(a + b)x^2 - ax + b = 0$

(vi) Find the cube roots of "-1".

(vii) Define ratio and give one example.

(viii) Find 'x' if 6 : x :: 3 : 5

(ix) If $y \propto x^2/z$ and y = 28 when x = 7, z = 2, then find y.

Q3. (Marks 12) Write short answers to any SIX parts.

(i) Resolve into partial fractions: 3/((x + 1)(x - 1))

(ii) Define an improper fraction.

(iii) If X = {1, 4, 7, 9} and Y = {2, 4, 5, 9} then find Y ∪ X

(iv) If A = {0, 2, 4} and B = {-1, 3}, then find B × A

(v) Find a and b if (3 - 2a, b - 1) = (a - 7, 2b + 5)

(vi) Define a bijective function.

(vii) Define cumulative frequency.

(viii) Define median.

(ix) Find arithmetic mean by direct method: 12, 14, 17, 20, 24, 29, 35, 45

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Define an angle.

(ii) Convert 15° to radians.

(iii) Find l, when $\theta = 180°$, r = 4.9cm

(iv) Simplify the expression to a single trigonometric function: $\tan x \cdot \sin x \cdot \sec x$

(v) Divide an arc of any length into two equal parts.

(vi) Define and draw the tangent to a circle.

> Note: Part - II, Attempt any THREE questions. Each question carries 08 marks. Question No. 9 is compulsory.

> حصہ دوم، کوئی سے تین سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 08 نمبر ہیں۔ سوال نمبر 9 لازمی ہے۔

Q5(a). (Marks 04) Solve the equation $7x^2 + 2x - 1 = 0$ by completing square.

Q5(b). (Marks 04) Prove that: $x^3 + y^3 = (x + y)(x + \omega y)(x + \omega^2 y)$

Q6(a). (Marks 04) Find x in the proportion: $p^2 + pq + q^2 : x :: (p^3 - q^3)/(p + q) : (p - q)^2$

Q6(b). (Marks 04) Resolve into partial fractions: $(3x + 7)/((x^2 + 1)(x + 3))$

Q7(a). (Marks 04) If U = {1, 2, 3, 4, ....., 10}, A = {1, 3, 5, 7, 9}, B = {1, 4, 7, 10} then verify that $(A \cap B)' = A' \cup B'$

Q7(b). (Marks 04) Find the standard deviation: 12, 6, 7, 3, 15, 10, 18, 5

Q8(a). (Marks 04) Verify the identity: $\cos^4\theta - \sin^4\theta = (\cos^2\theta - \sin^2\theta)$

Q8(b). (Marks 04) Circumscribe a regular hexagon about a circle of radius 3cm.

Q9. (Marks 08) Prove that one and only one circle can pass through three non-collinear points. -- OR -- Prove that the measure of a central angle of a minor arc of a circle, is doubled that of the angle subtended by the corresponding major arc.

## GROUP – II

# Mathematics (Objective) Group - II

**Time Allowed:** 20 Minutes | **Maximum Marks:** 15 | **Paper Code:** 7198

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. A set with no element is called:

(A) Sub set
(B) Empty set
(C) Singleton set
(D) Super set

ایسا سیٹ جس میں کوئی رکن نہ ہو کہلاتا ہے:

**(A)** تحتی سیٹ
**(B)** خالی سیٹ
**(C)** یک رکنی سیٹ
**(D)** سپر سیٹ

Q2. The different number of ways to describe a set is:

(A) 3
(B) 1
(C) 2
(D) 4

سیٹ کو بیان کرنے کے مختلف طریقوں کی تعداد ہوتی ہے:

Q3. A fraction in which the degree of numerator is less than the degree of the denominator is called:

(A) An equation
(B) An improper fraction
(C) An identity
(D) A proper fraction

کسر جس میں شمار کنندہ کی ڈگری مخرج کی ڈگری سے کم ہو کہلاتی ہے:

**(A)** مساوات
**(B)** غیر واجب کسر
**(C)** مماثلت
**(D)** واجب کسر

Q4. If a/b = c/d, then componendo property is:

(A) a/b = c/(c-d)
(B) a/(a+b) = c/(c+d)
(C) (a-b)/b = (c-d)/d
(D) ad/bc

اگر $a/b=c/d$ ہو تو ترکیب نسبت ہے:

Q5. The third proportional of $x^2$ and $y^2$ is:

(A) $y^2/x^2$ (B) $x^2y^2$

(C) $y^2/x^4$ (D) $y^4/x^2$

$x^2$ اور $y^2$ کا تیسرا متناسب ہے:

Q6. Product of cube roots of unit is:

(A) 0 (B) 1

(C) -1 (D) 3

اکائی کے جزر المکعب کا حاصل ضرب ہے:

Q7. The solution set of equation $4x^2 - 16 = 0$ is:

(A) {±4} (B) {±2}

(C) {4} (D) ±2

مساوات $4x^2 - 16 = 0$ کا حل سیٹ ہے:

Q8. The number of methods to solve a quadratic equation is:

(A) 1 (B) 2

(C) 3 (D) 4

دو درجی مساوات کو حل کرنے کے کتنے طریقے ہیں؟

Q9. The circumference of a circle is called:

(A) Chord (B) Segment

(C) Line segment (D) Boundary

دائرے کا محیط کہلاتا ہے:

**(A)** وتر **(B)** قطعہ

**(C)** خط قطعہ **(D)** سرحد

Q10. The arcs opposite to incongruent central angles of a circle are always:

(A) Congruent (B) Incongruent

(C) Parallel (D) Perpendicular

ایک دائرے میں غیر مماثل مرکزی زاویوں کے سامنے والے قوسین ہوتی ہیں:

**(A)** مماثل **(B)** غیر مماثل

**(C)** متوازی **(D)** عمودی

Q11. Two tangents drawn to a circle from a point outside it are _____ in length.

(A) Half
(B) Equal
(C) Double
(D) Triple

ایک دائرے کے بیرونی نقطے سے کھینچے گئے مماس لمبائی کے لحاظ سے ہوتے ہیں:

**(A)** نصف
**(B)** برابر
**(C)** دوگنا
**(D)** تین گنا

Q12. A complete circle is divided into:

(A) 180°
(B) 90°
(C) 120°
(D) 360°

مکمل دائرہ تقسیم کیا جاتا ہے:

Q13. 20° = _____

(A) 360'
(B) 630'
(C) 1200'
(D) 3600'

Q14. Mean is affected by change in:

(A) Scale
(B) Ratio
(C) Rate
(D) Place

حسابی اوسط _____ تبدیل کرنے سے اثر انداز ہوتا ہے۔

**(A)** پیمانہ
**(B)** نسبت
**(C)** شرح
**(D)** جگہ

Q15. A grouped frequency table is called:

(A) Data
(B) Frequency
(C) Frequency polygon
(D) Frequency distribution

گروہی تعددی جدول کہلاتا ہے:

**(A)** مواد
**(B)** تعددی
**(C)** تعددی کثیر الاضلاع
**(D)** تعددی تقسیم

## Mathematics (Subjective) Group - II

**Time Allowed:** 02:10 Hours | **Maximum Marks:** 60 | **Paper Code:** 7198

> Note: Part - I (حصہ اول)

Q2. (Marks 12) Write short answers to any SIX parts.

(i) [illegible in source scan]e: $x^2 + 2x - 2 = 0$

(ii) [illegible in source scan] in standard form: $x/(x+1) + (x+1)/x = 6$

(iii) [illegible in source scan]e exponential equation.

(iv) [illegible in source scan]te: $(1 - \omega - \omega^2)^7$

(v) [illegible in source scan] the quadratic equation having following roots: 1+i, 1-i

(vi) [illegible in source scan]thetic division to find the quotient and the remainder when: $(4x^3 - 5x + 15) \div (x + 3)$

(vii) [illegible in source scan] if 12, p and 3 are in continued proportion.

(viii) [illegible in source scan] inverse variation.

(ix) [illegible in source scan] $w \propto 1/v^2$ and $w = 2$ when $v = 3$, then find w.

Q3. (Marks 12) Write short answers to any SIX parts.

(i) [illegible in source scan]ve into partial fractions: $1/(x^2 - 1)$

(ii) [illegible in source scan]e an improper fraction.

(iii) [illegible in source scan] {1, 4, 7, 9} and Y = {2, 4, 5, 9} then find $Y \cup X$

(iv) [illegible in source scan] {0, 2, 4} and B = {-1, 3}, then find $A \times B$

(v) [illegible in source scan] a and b if $(3 - 2a, b - 1) = (a - 7, 2b + 5)$

(vi) [illegible in source scan]e a function.

(vii) [illegible in source scan]e cumulative frequency.

(viii) [illegible in source scan] three properties of arithmetic mean.

(ix) [illegible in source scan] arithmetic mean by direct method: 12, 14, 17, 20, 24, 29, 35, 45

Q4. (Marks 12) Write short answers to any SIX parts.

(i) [illegible in source scan]ert 225.75° to D° M' S'' form.

(ii) [illegible in source scan], when $\theta = 60°30'$, r = 15mm

(iii) [illegible in source scan] that: $(1 - \sin^2\theta)(1 + \tan^2\theta) = 1$

(iv) What is the length of the tangent?

(v) Define segment of a circle.

(vi) Define circumscribed circle.

(vii) Draw a tangent at any point on the circumference of a circle.

Note: Part - II, Attempt any THREE questions. Each question carries 08 marks. Question No. 9 is compulsory.

حصہ دوم، کوئی سے تین سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 08 نمبر ہیں۔ سوال نمبر 9 لازمی ہے۔

Q5(a). (Marks 04) Solve the equation using quadratic formula: $3/(x-6) - 4/(x-5) = 1$

Q5(b). (Marks 04) Find the value of k, if the roots of the equation are equal: $(3k + 2)x^2 - 5(k + 1)x + (2k + 3) = 0$

Q6(a). (Marks 04) Using the theorem of componendo-dividendo, find the value of $(x - 6a)/(x + 6a) - (x + 6b)/(x - 6b)$ if $x = 12ab/(a - b)$

Q6(b). (Marks 04) Resolve into partial fractions: $1/((x + 1)(x^2 + 1))$

Q7(a). (Marks 04) If U = {1, 2, 3, 4, ....., 10}, A = {1, 3, 5, 7, 9}, B = {1, 4, 7, 10} then verify that $A - B = A \cap B'$

Q7(b). (Marks 04) The marks of six students in Mathematics are as follows. Determine variance. [Students No.: 1, 2, 3, 4, 5, 6; Marks: 60, 70, 30, 90, 80, 42]

Q8(a). (Marks 04) If $\tan \theta = 4/3$, and $\sin \theta < 0$. Find the values of other trigonometric functions at $\theta$.

Q8(b). (Marks 04) Draw a circle which touches both the arms of angle 45°

Q9. (Marks 08) Prove that two chords of a circle which are equidistant from the center, are congruent. -- OR -- Prove that any two angles in the same segment of a circle are equal.